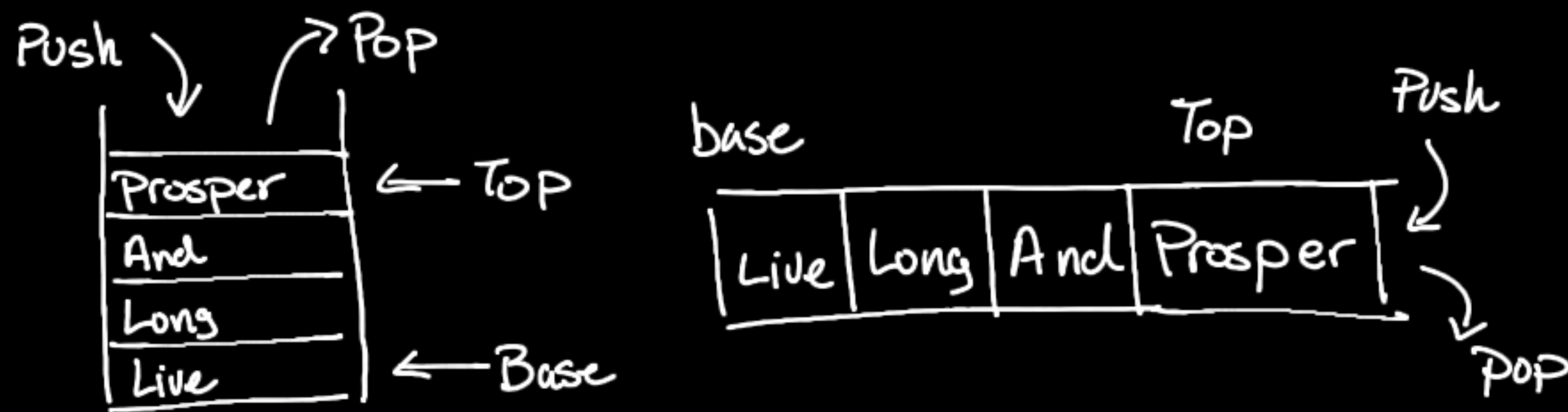
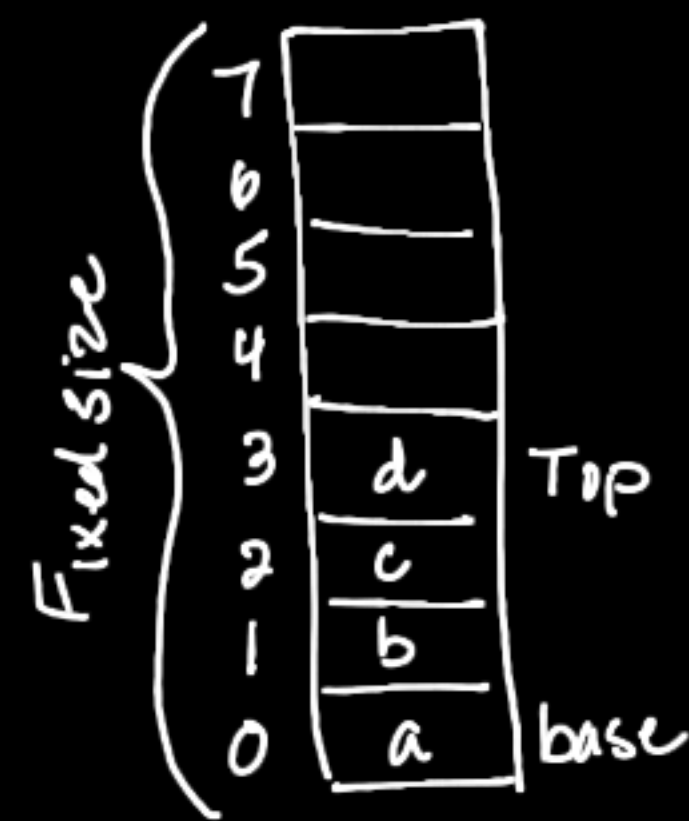


① Stack - LIFO



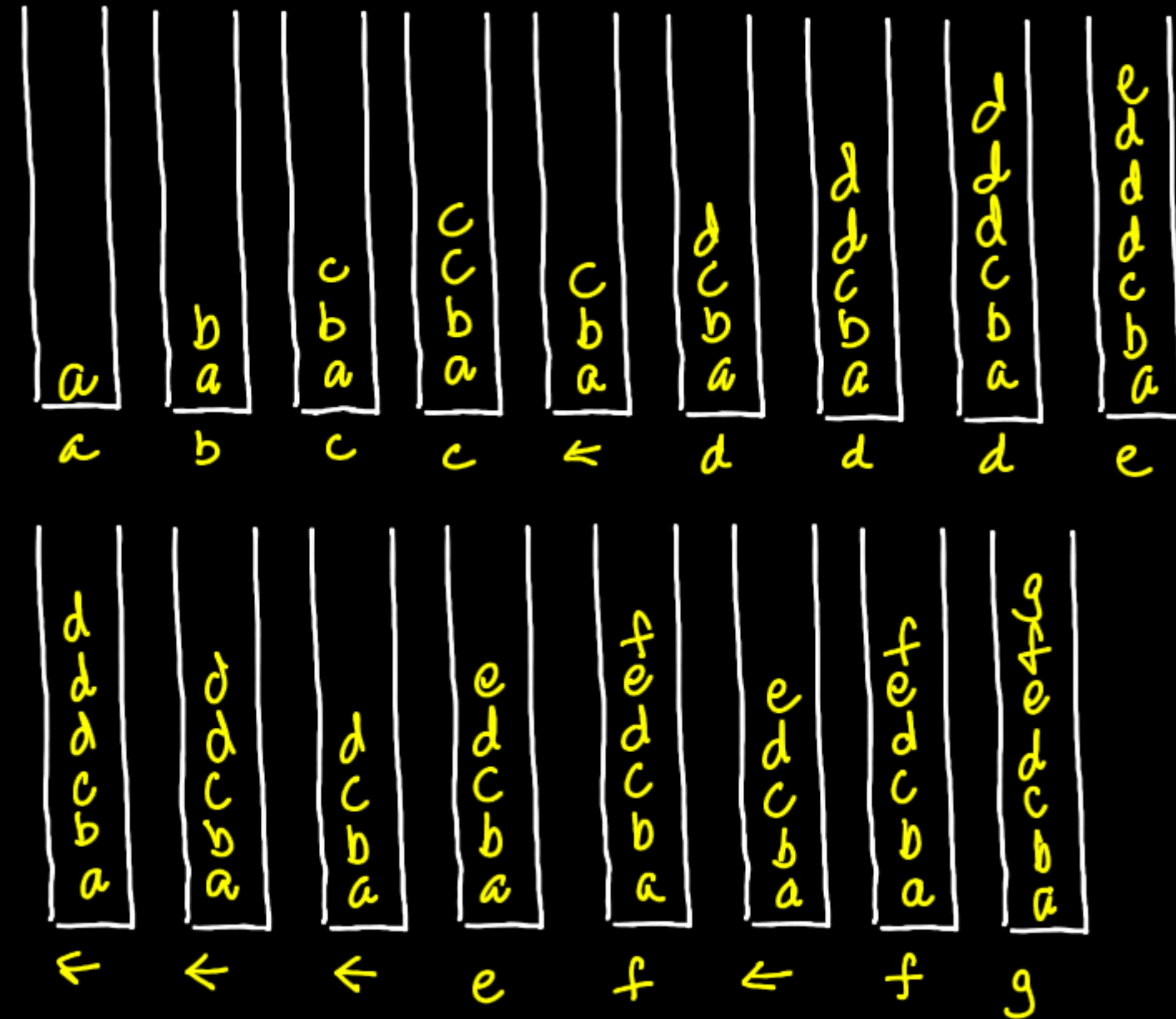
Array-based stack



Push(item)
if (Top + 1 > size - 1)
 print ("full stack")
else
 stack[Top + 1] = item
 Top = Top + 1

Pop()
if (Top < 0)
 print ("empty stack")
else
 item = stack[Top]
 Top = Top - 1
 return item

② Example Stack trace of $abcc \leftarrow ddd \leftarrow e \leftarrow f \leftarrow g$



③ Infix to Postfix expressions

Infix requires parentheses to control evaluation order

Post Fix does not because the operator comes after the operands

$$2 * (3 + 4) = 234 + * \\ \underbrace{\quad\quad\quad}_7 \quad \underbrace{\quad\quad\quad}_7 \\ 14 \quad 14$$

Post fix can be evaluated easily with a stack. Reading left to right, push operands, pop twice on operators, then perform operation and push result

$$10 \ 2 \ 8 \ * \ + \ 3 \ - = 23$$



Note: (2nd pop) (operator) (1st pop)

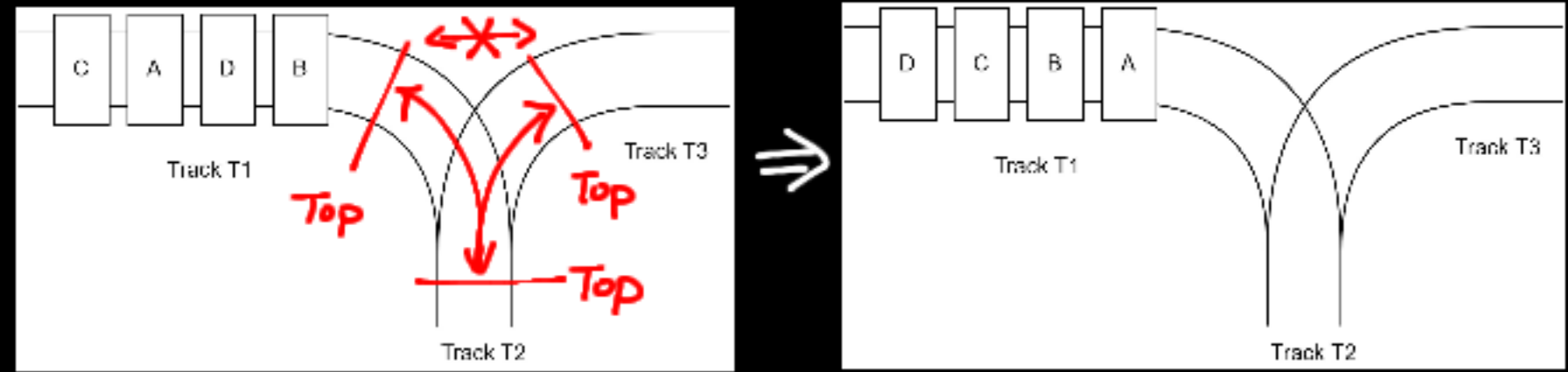
④ Convert infix to postfix

$$a - (b + c * d) / e$$

ch	Stack	Post fix
a	-	a
(	-(	a
b	-(	ab
+	-(+	ab
c	-(+	abc
*	-(+*	abc
d	-(+*	abcd
)	-(+	abcd*
	-(	abcd*+
/	-/	abcd*+
e	-/	abcd*+e
	-	abcd*+e/

- operands $\rightarrow$ post fix
- push (
- operator
if isEmpty, push operator
else
 pop while top is $\geq$ in precedence
 or stack is empty
 push operator
- if), pop operators until (
- if end of input, pop remaining

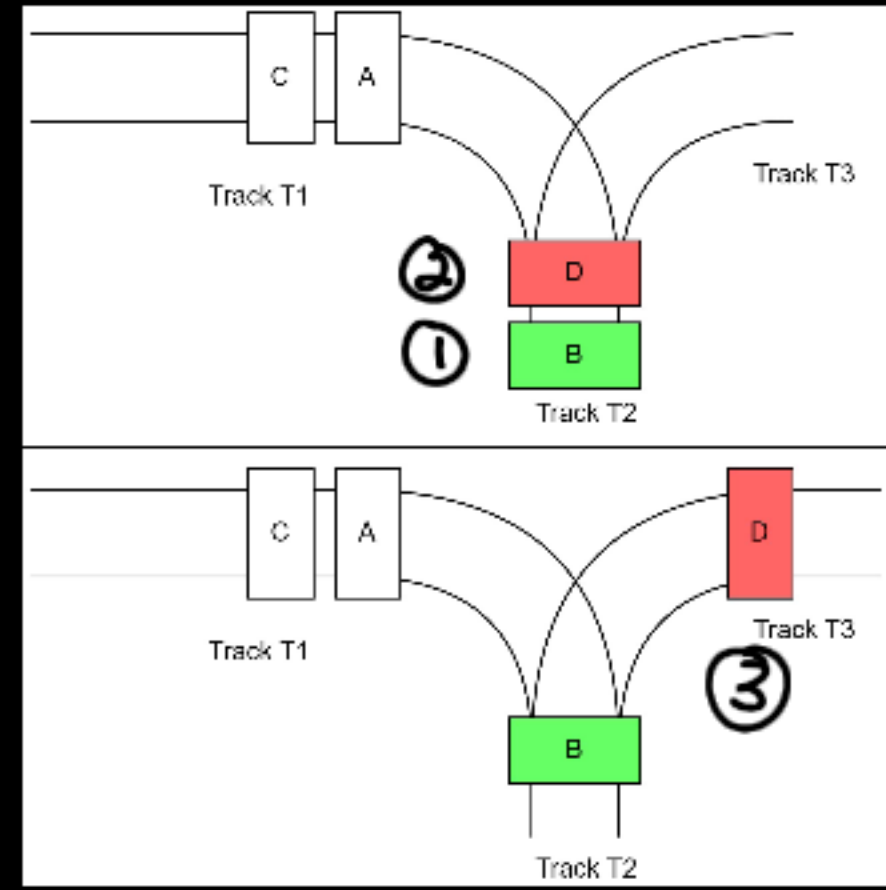
⑤ Train Car Sorting



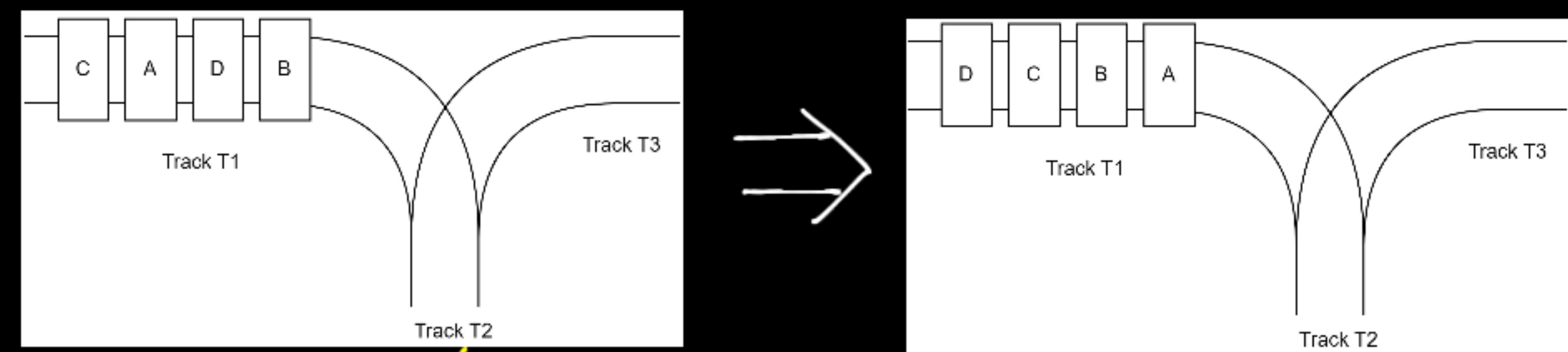
Representing

temp = T1.pop()
T2.push(temp)
or
T2.push(T1.pop())
~~T1.pop(B) NO!~~
~~T2.push(B) NO!~~
~~T1.push(pop(B)) NO!~~

- 1) T2.push(T1.pop())
B placed on T2
- 2) T2.push(T1.pop())
D placed on T2
- 3) T3.push(T2.pop())
D placed on T3



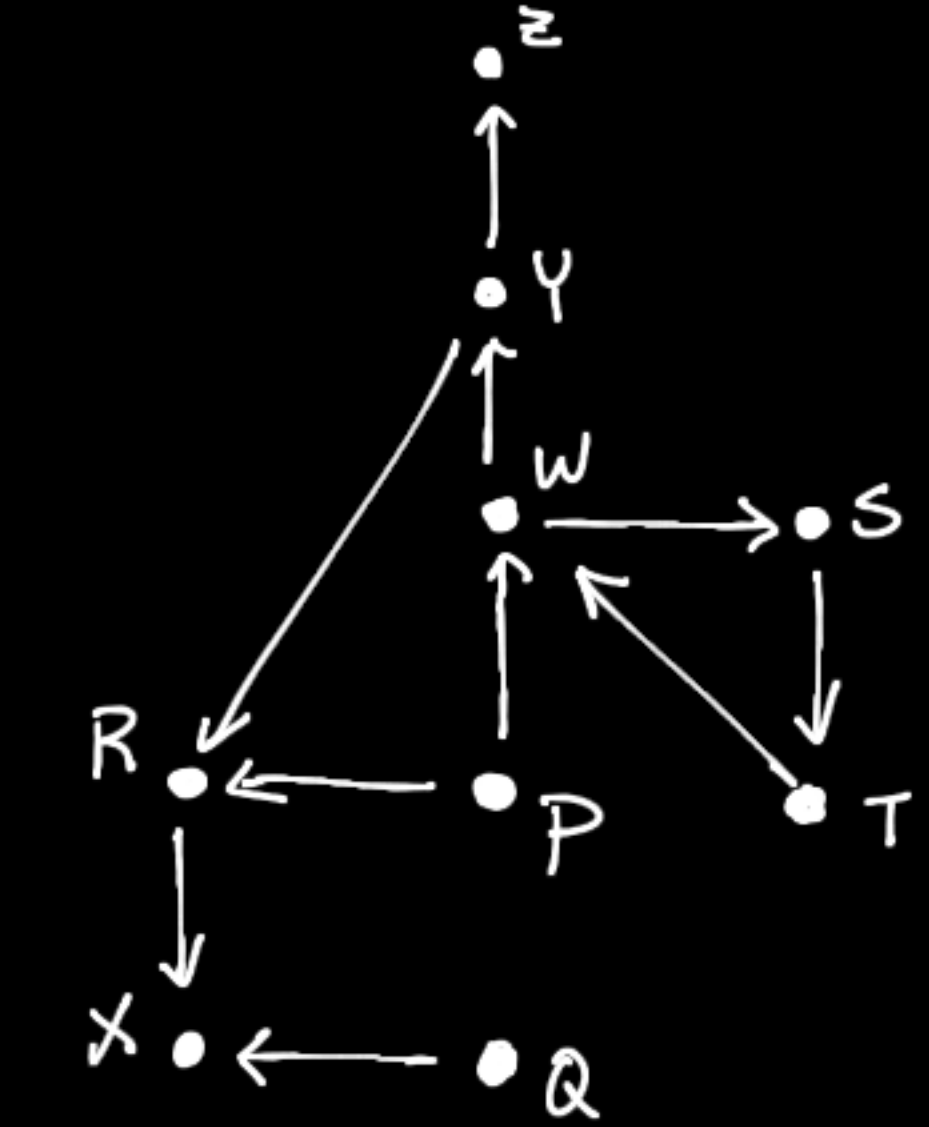
⑥ Train Car Sorting



- 1) T2.push(T1.pop()) ✓
- 2) T2.push(T1.pop()) ✓
- 3) T3.push(T2.pop()) ✓
- 4) T3.push(T2.pop()) ✓
- 5) T2.push(T1.pop()) ✓
- 6) T2.push(T3.pop()) ✓
- 7) T2.push(T1.pop()) ✓
- 8) T2.push(T3.pop()) ✓
- 9) T1.push(T2.pop()) ✓
- 10) T1.push(T2.pop()) ✓
- 11) T1.push(T2.pop()) ✓
- 12) T1.push(T2.pop()) ✓

T1 | ~~DCBA~~ DCBA
T2 | ~~BDACD~~
T3 | ~~DB~~

⑦ Challenge: SHIELD has many secret bases. Whenever a secret agent need to be picked up, the aircraft pilot wants the autopilot to find a route to the agent's location. For Security, there are limited routes from a location as seen in the directional graph.



- (1) Can only go in the arrow's direction
 - (2) Do not worry about return trips
 - (3) There may be unreachable locations
- Write an algorithm that will find possible routes from each secret base.
- Example - find routes from base P to each of the other bases.
- P → W
P → W → Y → Z
etc.